***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 13 Microbial Evolution and Systematics

13.1 Multiple Choice Questions

1) The earliest stromatolites were probably formed by
A) anoxygenic phototrophs.
B) anoxygenic lithotrophs.
C) oxygenic phototrophs.
D) oxygenic lithotrophs.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

2) What two gases were most abundant on early Earth?
A) O_2 and CO_2
B) N_2 and H_2
C) CO_2 and H_2
D) CO_2 and N_2
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

3) Compared with today, the temperature on Earth during its first half-billion years was probably
A) considerably warmer.
B) considerably colder.
C) about the same as today.
D) about the same as today on average, but the diurnal fluctuations were much greater.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

4) The earliest RNA probably functioned in
A) catalysis.
B) self-replication.
C) both catalysis and self-replication.
D) neither catalysis nor self-replication.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

5) Chemical reactions involving ________ have been proposed as energy-yielding reactions for the earliest organisms on Earth.
A) Fe, O_2, and H_2
B) S, O_2, and H_2
C) S, and O_2
D) S, and H_2
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

6) Evolution is driven by
A) random mutation.
B) novel metabolic pathways.
C) selection pressure.
D) selection pressure applied to random mutation.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.5

7) Microorganisms were probably restricted to the oceans and subsurface environments until
A) aquatic life brought them onto land.
B) chemoorganotrophy developed.
C) phototrophy evolved.
D) the ozone layer was made.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

8) Which of the following is an assumption used in the molecular clock approach?
A) Nucleotide changes accumulate in a sequence in proportion to time.
B) Nucleotide changes are generally NOT transferred to progeny.
C) Nucleotide changes are NOT random.
D) Nucleotide changes are subject to intense selective pressure.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.5

9) An important molecule in sequence-based evolutionary analyses of microbes are ________ genes.
A) ATPase
B) electron transport
C) SSU rRNA
D) tRNA
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.7

10) A monophyletic group is a group that
A) descended from one ancestor.
B) has the same fitness level.
C) possesses one taxonomic trait that is the same.
D) shares one phylogenetic marker.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.8

11) A gene for a specific trait may have more than one form, allowing the trait to vary. These sequence variants of a gene are called a(n)
A) alleles.
B) MLST.
C) horizontal gene transfers.
D) ribotypes.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.5

12) A key concept in evolution is that all mutations are
A) random.
B) neutral.
C) deleterious.
D) either deleterious or beneficial.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.5

13) LUCA is
A) the last universal common ancestor.
B) the individual ancestor of each of the three domains.
C) actually somewhat of a misnomer because it is now believed that each of the domains arose independently.
D) All of the answers are correct.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

14) Horizontal gene transfer
A) is so rare over evolutionary history that it is not considered when examining microbial evolution.
B) occurs within bacterial species.
C) complicates the construction of phylogenetic trees and the interpretation of specific traits in relation to evolution.
D) only affects the evolution of plasmids.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

15) Molecular sequencing suggests that mitochondria arose from a group of prokaryotic organisms within the
A) *Crenarchaeota*.
B) cyanobacteria.
C) iron-oxidizing bacteria.
D) *Alphaproteobacteria*.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.4

16) Which statement most closely expresses our present understanding?
A) The chloroplast is an ancestor of the cyanobacteria.
B) The cyanobacteria are descendents of the chloroplast.
C) The chloroplast arose from the incorporation of a cyanobacterial-like organism.
D) The chloroplast and the cyanobacteria are not closely (or specifically) related.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.4

17) Two tubes are inoculated from the same tube containing a bacterial culture. The cultures are then transferred every day for 2 months. All of the media and growth conditions are the same in every tube. After 2 months of cultivation, the fitness and genotype frequencies of the populations in the two tubes are compared. The fitness of the two cultures is the same, but the genotype frequencies are very different in the two cultures. How is this possible?
A) Two months is not long enough for different fitness levels to evolve even if the genotype frequencies change.
B) This result is not possible because different genotype frequencies would result in different fitness levels under the same growth conditions.
C) Natural selection caused the evolution of different genotype frequencies within the separate test tubes.
D) Genetic drift within the small populations in the test tubes resulted in different genotype frequencies.
Answer: D
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 13.5

18) If you allowed 10 identical parallel *Salmonella* cultures to evolve for 10,000 generations under new growth conditions with very little nitrogen, the parallel cultures would
A) evolve identical adaptations to use the nitrogen source provided in the media.
B) not change or adapt very much over this small number of generations.
C) each accumulate different random mutations resulting in different adaptations to use the nitrogen in the media.
D) direct mutations to occur in nitrogen utilization and uptake genes in order to adapt rapidly to the culture conditions.
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 13.5

19) Both deletions and insertions occur during the evolution of microbial genomes. Insertions bring in new genes that may be useful for the cell. Deletions
A) may increase fitness of a microorganism by eliminating unneeded genes.
B) always result in a severe loss of fitness for the microorganism.
C) are uncommon because they are usually lethal.
D) always result in a severe loss of fitness but keep microbial genomes compact.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.6

20) Polyphasic taxonomy uses methods that include
A) phenotypic methods.
B) genotypic methods.
C) phylogenetic methods.
D) phenotypic, genotypic, and phylogenetic methods.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.6

21) Ribotyping
A) bypasses sequencing and sequence alignments.
B) exploits unique DNA restriction patterns.
C) allows discrimination between species and different strains of a species.
D) bypasses sequencing and sequence alignments, exploits unique DNA restriction patterns, and allows discrimination between species and different strains of a species.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.9

22) Which is NOT a characteristic of a ″primitive″ state of microbial evolution?
A) hyperthermophilic
B) aerobic metabolism
C) small genome
D) branching near the root of the evolutionary tree of life
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.1

23) Because DNA-DNA hybridization reveals subtle differences in genes, it is useful for differentiating organisms from different
A) domains.
B) families.
C) orders.
D) strains or species.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.8

24) Which taxonomic tool would scientists use if they wanted to determine if an outbreak of food poisoning was caused by a particular strain of a pathogen?
A) fluorescence *in situ* hybridization
B) DNA-DNA hybridization
C) multilocus sequence typing
D) fatty acid methyl ester analysis
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.9

25) It is believed that phototrophy arose approximately 3.3 billion years ago in
A) *Bacteria*.
B) *Archaea*.
C) *Eukarya*.
D) LUCA.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

26) The oxic atmosphere created conditions that led to the evolution of various new metabolic pathways, such as
A) sulfide oxidation.
B) nitrification.
C) iron oxidation.
D) sulfide oxidation, nitrification, and iron oxidation.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.2

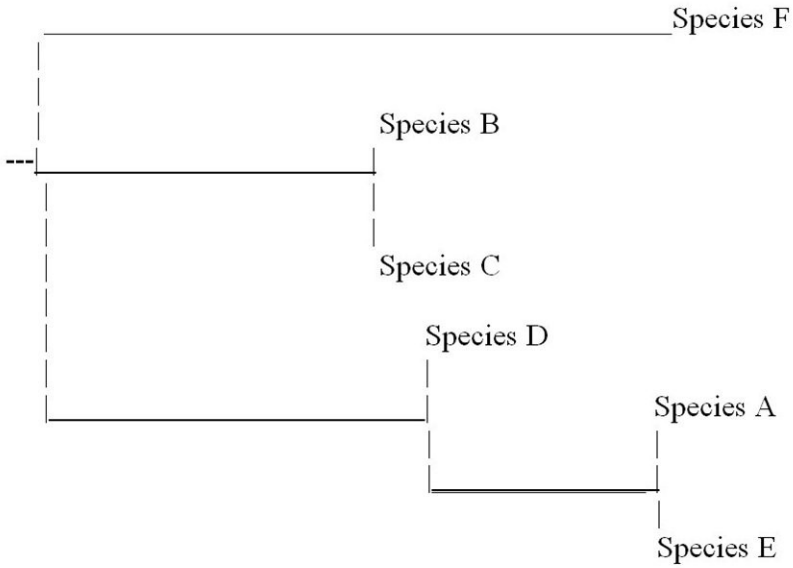
27) You are studying 12 new isolates from the human skin. Their average nucleotide identity for shared orthologous genes is 97%.The isolates would most likely be
A) classified as individual strains of the same species.
B) classified as individual species of the same genus.
C) split into different families.
D) classified as the same species if they can mate via conjugation.
Answer: A
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 13.8

28) The pan genome of a microbial species is constantly changing because of
A) bottleneck events.
B) horizontal gene transfer.
C) substitutions.
D) substitution and bottleneck events.
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.6

29) Microbial ________ studies the diversity of microorganisms and links their phylogeny with ________.
A) taxonomy / genotype
B) phylogeny / phenotype
C) taxonomy / phenotype
D) systematics / taxonomy
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.7

30) The earliest photosynthetic microbes, before the cyanobacterial lineage developed, oxidized substances other than water. What was produced by these microbes instead of oxygen?
A) nitrate
B) elemental sulfur
C) ferrous iron
D) hydrogen sulfide
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

31) Based on the phylogenetic tree below, which of the following statements is FALSE?



A) Species E is more closely related to species A than to species D.
B) Species D is more closely related to species C than to species E.
C) Species F is more closely related to species D than to species E.
D) Species C is more closely related to species B than to species D.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.7

32) The early energy reactions used hydrogen, which is a powerful
A) oxidant.
B) electron donor.
C) electron acceptor.
D) stabilizer.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

33) Oxygen did not accumulate in the early atmosphere until it reacted with reduced materials, especially ________, in the oceans.
A) hydrogen
B) elemental sulfur
C) ferrous iron
D) nitrate
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

34) As oxygen appeared in the atmosphere, ________ also accumulated, which formed a protective barrier that protects the Earth from ________.
A) ozone / UV radiation
B) elemental sulfur / volcanism
C) ferrous iron / hydroxylating radicals
D) sulfate / hydrogen sulfide
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

35) Two eukaryotic organelles that are hypothesized to be the result of endosymbiosis are the ________ and the ________.
A) nucleus / Golgi body
B) mitochondrion / chloroplast
C) endoplasmic reticulum / Golgi body
D) hydrogenosome / chloroplast
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.4

36) Microbial species have a core genome and a pan genome. What is the difference between the two?
A) The core genome consists of all the nucleic acid polymerases and translation enzymes, while the pan genome consists of all the biosynthetic pathways.
B) The core genome is the set of genes within the mitochondria, while the pan genome is the set of genes in the nucleus of a species.
C) The core genome is a set of genes shared by all members of a species, while the pan genome includes the core genes as well as genes that are not shared by all members.
D) The core genome is the set of genes introduced by horizontal gene transfer, while the pan genome is the set of genes that is not transferred horizontally.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.6

37) What characteristics make a gene a good candidate for determining the evolutionary relationships between organisms?
A) highly conserved
B) universally distributed
C) transferred horizontally between species
D) highly conserved and universally distributed
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.7

38) Which of the following statements is FALSE concerning the universal tree of life?
A) Previous versions of the universal tree of life were based largely on fossils and comparative biology, which overlooked the diversity of and relationships between most prokaryotes.
B) The current universal tree of life is supported by multiple genes including SSU rRNA sequences.
C) The current universal tree of life depicts three domains of life, two of which are prokaryotic.
D) The current universal tree of life is based on molecular sequences and completely changes our view of evolutionary relationships between eukaryotic species.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.3

39) ________ formed the semipermeable membrane-like surfaces for the earliest life forms.
A) Proteins
B) Hydrogen sulfide
C) Lipid bilayers
D) RNA
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.4

40) The first catalytic and self-replication biological molecule was most likely
A) RNA.
B) DNA.
C) proteins.
D) ATP.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

41) The evolutionary history of a group of organisms is called its ________ and it is inferred from ________.
A) taxonomy / phenotype
B) phylogeny / nucleotide sequence data
C) phylogeny / phenotype
D) taxonomy / morphology
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

42) One widely used method of examining evolutionary relationships is ________, which is based on the assumption that evolution is most likely to have proceeded by the path requiring fewest changes.
A) topology
B) neighbor joining
C) parsimony
D) systematics
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.7

43) Multilocus sequence typing (MLST) involves sequencing several different
A) genomes.
B) housekeeping genes.
C) tRNA genes.
D) rRNA genes.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.9

44) Systematic analysis now commonly includes ________ to identify, characterize, and determine relationships between new strains of bacterial species.
A) whole genome analysis
B) microscopy
C) staining
D) pigments
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.9

13.2   True/False Questions

1) The atmosphere of primitive Earth is usually described as an oxidizing atmosphere.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

2) The earliest nucleic acid was probably a simple DNA molecule.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

3) The establishment of DNA as the genome of the cell may have resulted from the need to store genetic information in a more stable form than RNA.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

4) Eukaryotes originated after the rise in atmospheric oxygen.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.4

5) A type strain is the first strain of a new species to be identified and described.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.10

6) Sequencing technology and molecular phylogenetic analyses have had very little impact on our understanding of the evolution and diversity of life on Earth.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.9

7) Molecular phylogeny and rRNA analysis provided the evidence used to separate *Bacteria* and *Archaea* into distinct domains.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

8) Organisms with greater phylogenetic distance in their genomes have less gene exchange than those with less phylogenetic distance.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.6

9) Microbial species are difficult to define because they are seldom monophyletic.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.8

10) Organisms within a species should have strong phenotypic cohesiveness.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.8

11) At present there are four phyla in the domain *Bacteria*.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

12) The universal phylogenetic tree shows that the Bacteria diverged from the Archaea before the Eukarya diverged from the Archaea.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

13) The 16S rRNA gene sequence is an approximately 1,465 bp linear strand of single stranded RNA.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.7

14) DNA-DNA hybridization is a sensitive method for revealing subtle genetic differences because it measures the degree of sequence similarity between two genomes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.8

15) One phenotypic trait used for species identification and description is the analysis of the types and proportions of the fatty acids present in the cytoplasmic and outer membranes. The methodology employed is often nicknamed FAME.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.9

16) In taxonomy, family is a more general term than order.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.10

17) Oxygen was a driving factor in the formation of eukaryotic cells.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

18) The primary domains were founded based on comparative ribosomal RNA gene sequencing.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

19) All unicellular organisms belong to the same domain of life.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

20) Approximately 3.7 billion years ago *Archaea* and *Bacteria* diverged as being distinct from each other.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.3

21) When exposed to UV light, oxygen gas produces ozone gas.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.2

22) Horizontal gene transfer is one plausible explanation as to why organisms in *Archaea*, *Bacteria*, and
Eukarya still share so many genes among such distinct domains.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.3

13.3   Essay Questions

1) Explain the endosymbiotic theory. What is the evolutionary value of endosymbiosis?
Answer: The theory states that a chemoorganotrophic bacterium and a cyanobacterium were stably incorporated into a cell and gave rise to a mitochondrion and chloroplast (respectively), which formed a eukaryotic cell. Endosymbiosis provides a mutualistic relationship where the incorporated cells are protected from direct contact with the environment, and the cell that incorporated the cells gains metabolic diversity.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.4

2) Describe the hydrogen hypothesis and why it is favored over other hypotheses.
Answer: The hydrogen hypothesis states that a hydrogen-consuming archaea served as a host for the stable incorporation of a hydrogen-producing facultative anaerobic bacterium. This hypothesis describes the first eukaryotic cells as ones without nuclei. It accounts for eukaryotes having bacteria-like ATP-producing pathways in hydrogenosomes and the presence of bacteria-like lipids in eukaryotes. Neither of these is accounted for in other hypotheses that state that eukaryotes first contained nuclei and lacked mitochondria.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 13.4

3) Carl Woese was an important figure in microbial classification and changed our understanding of the evolution and diversity of life. What breakthrough did he make that allowed us to infer evolutionary relationships between organisms? What assumptions was his breakthrough based on?
Answer: During the 1970s, Dr. Carl Woese pioneered the usage of SSU rRNA genes for phylogenetic analysis. He realized that SSU rRNA genes could be used to infer evolutionary relationships because they were (1) universally distributed, (2) have the same function in every organism, (3) change slowly over evolutionary time (highly conserved), and (4) of adequate length to provide sufficient information about very deep, or old, evolutionary relationships. He assumed that changes in DNA sequence are reliably passed on to offspring (they are heritable) and that nucleotide changes between two organisms will accumulate over time since they last shared a common ancestor. Homologous sequence comparison thus allows us to infer evolutionary relationships between nucleotide sequences. Genes that are not often exchanged through horizontal gene transfer (such as SSU rRNA or other housekeeping genes) can be used to infer evolutionary relationships between organisms.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 13.3

4) Why is the biological species concept useful for *Eukarya* but not meaningful for *Archaea* and *Bacteria*? How are prokaryotic species defined?
Answer: A species is an interbreeding population of organisms that is isolated from other interbreeding populations, and this definition generally works well in defining eukaryotic species that reproduce sexually. However, prokaryotes undergo asexual reproduction and are subject to horizontal gene transfer in the environment, which complicates the definition of a species in *Archaea* and *Bacteria*. Prokaryotic species are defined as a monophyletic group of strains that share important phenotypic traits and have relatively high genomic and SSU rRNA sequence similarity. (Answers could include the accepted values of greater than 70% DNA-DNA hybridization and 97% SSU rRNA sequence similarity but the concept of similarity of a specific level and a monophyletic origin is most important.)
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 13.8

5) You have isolated a microbe from an environmental sample. The microbe has the ability to perform a new metabolic reaction at a very low temperature, so you are excited that it could be a new species. What experiments should you perform to determine if your isolate is truly a new species? What process will you follow to officially name your isolate?
Answer: Experiments to determine the isolate′s characteristics and distinguishing traits must be performed which, among others, involve a series of biochemical and physiological tests. The morphology and cell wall structure should also be determined. To define it as a species and determine its relationship to other organisms the SSU rRNA and DNA. DNA hybridization values in comparison to closely related organisms must be determined. If the isolate is sufficiently different, viable cultures must be sent to at least two international culture collections. A manuscript can then be published describing the organism and a proposed genus and species name.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.9

6) *Rhodobacter* cells perform photosynthesis in the presence of light and grow heterotrophically in the dark. After being cultured in total darkness for multiple generations, phenotypic and genotypic changes occur. What phenotypic changes occur and what evolutionary processes are driving them?
Answer: When a population of phototrophs is grown in the dark for multiple generations, the population will tend to lose the ability to produce photopigments and may even lose the ability to perform photosynthesis altogether. This occurs because there is no selection for the production of the pigments and enzymes necessary for photosynthesis. The production of photosynthetic pigments and enzymes is metabolically expensive, thus producing them when they are not needed results in a competitive disadvantage in the dark cultures, thus random mutants that produce fewer pigments take over *Rhodobacter* cultures grown in the dark. The evolutionary processes behind this phenotypic change are random mutations to the genome, followed by the selection of mutants that grow better in the dark.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.5

7) How did cyanobacteria, oxygen, and ozone impact the evolution of eukaryotic cells?
Answer: The first phototrophs on Earth were anoxygenic bacteria, meaning that oxygen was NOT produced during photosynthesis. Eventually the *Cyanobacteria* evolved and oxygenic photosynthesis began to produce oxygen—the first major source of oxygen on Earth. As the oxygen was produced, it first reacted with all of the reduced chemicals found on early Earth, but eventually it began to accumulate in the atmosphere. As the oxygen gas accumulated, UV light converted the oxygen into ozone. Ozone absorbs UV irradiation and the formation of the ozone layer protected the Earth′s surface from strong UV irradiation. Because UV irradiation damages DNA and other biomolecules, before the ozone layer life could only exist in protected environments, such as deep in the ocean or in the subsurface. The production of oxygen by *Cyanobacteria* thus did two very important things that may have helped eukaryotic and multicellular organisms evolve. The oxygen was a powerful oxidant and excellent source of energy (excellent electron acceptor for respiration) and the large majority of eukaryotic cells use aerobic respiration to generate large amounts of energy for cellular growth. All of this energy may have fueled the growth and diversity of multicellular organisms. In addition, the protection of the ozone layer allowed life to expand into new environments.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 13.2

8) Life is believed to have originated at hydrothermal springs on the ocean floor rather than on or near the surface of Earth. What conditions made the hydrothermal springs the likely place for the beginning of life? Relate the conditions of the early oceans to the likely metabolism of the first cells.
Answer: Aside from the temperature, the environmental conditions of Earth would have been less hostile for life in hydrothermal springs than on land where exposure to UV light would have destroyed biological molecules. The springs also likely provided a steady supply of energy, such as H_2 and H_2S, and where the hot water interfaced with colder water likely served to form precipitates that eventually became life. The metabolism of the first cells did not depend on oxygen, as the atmosphere lacked molecular oxygen, but probably utilized electron transfer reactions between hydrogen, hydrogen sulfide, and reduced iron to produce energy.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.1

9) How does the Black Queen Hypothesis explain the effect of gene deletions on bacterial evolution?
Answer: The Black Queen Hypothesis states that some organisms can increase their fitness by selection loss (deletion) of specific genes. The loss of the genes decreases the genome and proteome size and thus the amount of energy it takes to reproduce, increasing the organism′s fitness. However, it also makes the organism dependent on other members of the community to provide the functions or gene products that have been deleted. The dependent organisms that have lost genes will only be competitive as long as others in the community provide the functions/products that have been lost. Thus in environments with a stable community composition, organism may evolve smaller genomes through gene deletion. This has been observed in the open ocean and in some symbiotic organisms.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.6

10) Explain the strengths and limitations of using 16S rRNA for phylogenetic analyses.
Answer: Due to function constraints, the 16S rRNA gene sequence has changed relatively little over time and is therefore an excellent individual nucleotide sequence useful in phylogenetic studies. It is not especially good at distinguishing among different species within a genus, because there is often not enough variation between the sequences. There are large and freely available databases dedicated to cataloging 16S rDNA sequences, and it remains the ″gold standard″ today for phylogenetic analyses.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 13.7

11) Can gene frequencies change in the absence of selection? Why or why not?
Answer: Gene frequencies do change in the absence of selection. Genetic drift is a random process through which gene frequencies change. By chance, some members of a population will have more offspring than others, while other members may be killed or inhibited by CHANCE, not selection. An example is the random selection of very few cells from one test tube culture to another test tube. Which cells are transferred will affect which genes are passed on to offspring, but the cells transferred were random and unrelated to the fitness of the cells. Thus gene frequencies can change (drift) randomly over time even when there is no selection on that particular gene. Although genetic drift is separate from selection, it is still an important part of evolution.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.5

12) What type of methods are used in microbial systematics? Why are multiple methods used to characterize and classify microbial species? Be sure to define the methods and describe the general strengths and weaknesses of each type.
Answer: Microbial systematics is polyphasic and uses three types of methods to identify and describe microbial species: (1) phenotypic — the observed characteristics of an organism, (2) genotypic — the genetic features of an organism, and (3) phylogenetic — the evolutionary relationship of an organism to other organisms. Phenotypic methods are important because they describe the important traits of an organism that define the organism′s function and impact on humans or the environment. However, many phenotypic traits are not unique to a species, so identifying the huge number of microbial species based only on phenotypic traits is not feasible. Genetic methods complement phenotypic methods and help us quickly characterize both gene content and genetic similarity to other organisms. Lastly, phylogenetic methods help to organize microbial species in a way that is congruent with our understanding of evolution, and thus give both structure and meaning to the taxonomy of microorganisms.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 13.9